

1 Scientific/Technical/Management Plan

Under the current paradigm, the majority of the prime mission time for NASA’s Habitable Worlds Observatory (HWO) would be spent on an exoplanet direct imaging survey. Overall survey time is driven, in large part, by the need to repeatedly revisit systems to confirm if detected sources orbit within their host’s habitable zone. The work proposed below aims to dramatically reduce these revisit demands by developing a novel detection follow-up strategy that maximizes the information gleaned from astrometric and photometric detections of HWO targets. These gains would reduce the overall survey duration while increasing the exo-Earth haul, and our proposed studies come at a time during HWO’s development where telescope size—the major cost driver—is being weighed against exoplanet yield.

1.1 Introduction, Goals, and Significance

The Astro2020 Decadal Survey [16] charge regarding the deliverable from a direct imaging survey of exoplanets in reflected light is simple to state yet wonderfully challenging in its interpretation and implementation: “a robust sample of ~ 25 atmospheric spectra of potentially habitable exoplanets.” The under-development NASA HWO [4]—a 6-meter-class, coronagraph-equipped space-based observatory that would obtain ultraviolet (UV), optical, and near-infrared (NIR) reflected-light spectra of exoplanets orbiting nearby Sun-like stars—will meet the Decadal charge and deliver to the community yet another pivotal Great Observatory. The opening of a NASA Project Office for HWO in 2025 plainly signals that **now is the time for HWO science maturation, technology development, and observatory architectural trades.**

However, the Decadal goal of “ ~ 25 atmospheric spectra of potentially habitable exoplanets” sets up multiple tensions. Perhaps the most obvious tension is between exoplanet science and the other transformational science that HWO can deliver across solar system studies, stellar astronomy, galactic astrophysics, and cosmology. If the requisite survey time for potentially Earth-like planets is too long, then all other science areas can suffer. A less obvious tension is between detection and characterization—the **more time it takes for HWO to find small worlds orbiting in their host star’s habitable zone the less time that is available for characterizing these worlds** (Figure 1).

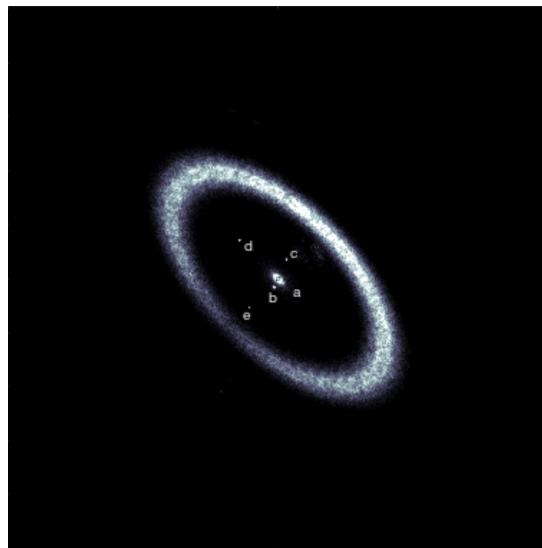


Figure 1: Simulated observation of an exoplanet system (HabEx Final Report, their Figure 3.3-2.) Takeaway: The execution of a HWO-like mission requires knowing if any sources in an image like this are key follow-up targets.